Unpacking Household Budgeting Strategies through a Transportation Lens

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Abstract
This is where we will put our abstract.

Plain Language Summary

This is a plain language summary

Source: Article Notebook

1 Introduction

Statistics I can point out

- The amount of budget allocated to transportation is more than it was a century ago (Johnson et al., 2001) but, in the past few decades it hasn't changed much (Bureau of Transportation Statistics, 2024)
- How much does the average household spend on transportaiton? I can probably get some good data from Bureau of Transportation Statistics (2024)
  - rural averages about 2.5 vehicles per household and urban averages about 1.7
  - transportation share has gone down since 2000, but has generally stayed somewhere between the 15 - 20 percent range. That doesn't mean people are spending less on transportation. It more likely means that people are spending more in other categories.
  - The number of earners in a household is closely related to how much is spent on transportation.
  - Since 1960, a larger percent of US households have at least one car. It was in the 80s when the percentage of 2-car households surpassed the percentage of 1-car households
- (Mattson, 2020) Car ownership, family size, income, housing type, and other location factors are connected with transportation expenditures.
  - People with bigger families tend to spend more on transportation
  - People in single-family homes also generally spend more on transportation
- Lower income households spend a bigger percent of their budget on transportation (transportaiton burdened) (Molloy et al., 2024)

This is my start at trying to talk about the statistics

A major portion of a household's expenditures is allocated to transportation. The amount people spend on transportation has clearly increased in the last century, but has averaged around 15 - 20 percent over the last few decades. In 2023, the average household spent approximately 17 percent of their budget on transportation expenses (Bureau of Transportation Statistics, 2024; Johnson et al., 2001).

The percent of budget spent on transportation varies widely from household to household, and there are many factors that play a role in a household's transportation expenditures. Some possible factors include income, number of vehicles, number of workers, number of children, and location.

Lower income households seem to be more burdened by transportation expenses as it often takes up a larger portion of their overall expenses (Molloy et al., 2024).

Larger family sizes and living in single-family homes both tend to be correlated with increased transportation spending (Mattson, 2020).

Car ownership has been increasing, with less than 10 percent of carless households in 2023 compared to over 20 percent in 1960. Before 1980, there was a larger portion of one-car households compared to two-car households. Since then, the number of two-car households has surpassed the number of one-car households.

This is what Katie Wrote

Households juggle how to allocate their budgets: whether to invest in a reliable car, pay for quality childcare, secure housing in a good school district, or set money aside for leisure. These everyday choices shape how families live and move, reflecting the trade-offs they make to balance competing priorities. Transportation often sits at the center of these decisions, not only because it can be a significant expense, but also because choosing to buy and maintain a car versus relying on public transit represents a long-term commitment and a broader lifestyle choice. Its relative weight compared to housing, childcare, and other spending varies widely across families. The relationship between household budgeting and mobility is shaped not only by causal direction but also by how families prioritize and weight different needs. On one hand, mobility resources such as car ownership can structure the household budget: households with no or only one vehicle may spend far less on transportation, freeing up income for other essential or discretionary categories. On the other hand, underlying family structures and preferences can drive budget allocation choices that, in turn, shape transportation behavior. Larger families may prioritize childcare or invest in higher-quality housing in areas with better schools, limiting what remains for transportation. Others may emphasize frugality across all categories or deliberately substitute toward lower-cost transit options. Understanding both the direction of influence and the weight assigned to different budget categories is critical for transportation planning and policy, as these dynamics reveal how families navigate competing priorities under varying demographic and mobility contexts.

The purpose of this research is to explore how household budgets are structured around transportation decisions and how this impacts other spending categories. Using the Consumer Expenditure Survey (CEX), we will perform a Latent Class Analysis (LCA) to find groupings based on a household's transportation expenses. These groupings can help us find groups of spenders with similar patterns to help us predict transportation expenses based on the household's characteristics.

2 Literature Review

Source: Article Notebook

Table 1 summarizes the literature that was reviewed for this study. As seen in the table, there were a total of 35 studies reviewed with 19 studies that use the consumer expenditure survey.

Table 1: Summary of the Literature

Study	Use Consumer Expenditure Survey Data	Household Choices	Transportation	Budgets	Transportation and Budgets
Morris & Wigan (1979)	NA	NA	NA	NA	1
Skinner (1985)	1	NA	NA	1	NA
Nayga (1998)	1	NA	NA	1	NA
Blumenberg (2003)	1	NA	NA	NA	1
Thakuriah & Liao (2005)	1	NA	NA	NA	1
Hong et al. (2005)	1	NA	NA	NA	1
Thakuriah (Vonu) & Liao (2006)	1	NA	NA	NA	1

Table 1: Summary of the Literature

Study	Use Consumer Expenditure Survey Data	Household Choices	Transportation	Budgets	Transportation and Budgets
Fontes & Fan (2006)	1	NA	NA	1	NA
Choo et al. (2007)	1	NA	NA	NA	1
Rachel B. Copperman & Bhat (2007a)	NA	NA	1	NA	NA
Sener & Bhat (2007)	NA	1	NA	NA	NA
Rachel B. Copperman & Bhat (2007b)	NA	1	NA	NA	NA
Haas et al. (2008)	1	NA	NA	NA	1
Sener et al. (2008)	NA	1	NA	NA	NA
Ferdous et al. (2010)	1	NA	NA	NA	1
Souche (2010)	NA	NA	1	NA	NA
Paleti et al. (2011)	NA	1	NA	NA	NA
Sabelhaus et al. (2013)	1	NA	NA	1	NA
Deka (2015)	1	NA	NA	NA	1
Bernardo et al. (2015)	NA	1	NA	NA	NA
Lino et al. (2017)	1	NA	NA	1	NA
McCarthy et al. (2017)	NA	NA	1	NA	NA
Mattson & Peterson (2019)	1	NA	NA	NA	1
Leung et al. (2019)	NA	1	NA	NA	NA
Mattson (2020)	1	NA	NA	NA	1
Osborne et al. (2021)	1	NA	NA	1	NA
Hastings (2022)	1	NA	NA	1	NA
Lu et al. (2022)	NA	NA	1	NA	NA
Amirnazmiafshar & Diana (2022)	NA	NA	1	NA	NA
Duncan et al. (2023)	NA	NA	NA	1	NA
Molloy et al. (2024)	1	NA	NA	NA	1
Bureau of Transportation Statistics (2024)	1	NA	NA	NA	1
Hargunani et al. (2024)	NA	NA	NA	1	NA
Klein (2024)	NA	NA	1	NA	NA
Bilgin et al. (2025)	NA	NA	1	NA	NA

Source: Lit Review Table

The literature relating to this study has been classified into four groups: (1) Household Choices and Activity Patterns, (2) Household Transportation Choices, (3) Household Spending and Budgets, and (4) Household Transportation Budgets. The following sections describe the relevant findings from literature in each of these groups.

2.1 Household Activities and Transportation Choices

The literature shows that household dynamics affect the activity patterns of those in the household, and many studies focus on the role children play in these household activities. Some studies aim to find how the presence of children affect the activities of the household (Bernardo et al., 2015), and many studies focus on how children's activities are affected by household demographics (Rachel B. Copperman & Bhat, 2007b;

Leung et al., 2019; Paleti et al., 2011; Sener et al., 2008; see Sener & Bhat, 2007). A common finding in these studies is that child activity participation is associated with the environment the child lives in along with household income, parent activities, and other household demographics.

Among the studies on household choices is a group of literature with a specific focus on a household's transportation decisions. A common finding in this group of literature is that household composition plays a role in its transportation choices. Some studies in this group are concerned with the way households use cars or the effect of car ownership on transportation decisions (Amirnazmiafshar & Diana, 2022; Bilgin et al., 2025; Klein, 2024; Lu et al., 2022). These studies show there is a major connection between car ownership and transportation decisions. Other studies are focused on travel choices more generally (Rachel B. Copperman & Bhat, 2007a; Souche, 2010). For an extensive literature review on the transportation choices of households with young children, see the study done by McCarthy et al. (2017).

2.2 Household Spending and Budgets

Another set of studies focuses on household budgets and household spending patterns (Fontes & Fan, 2006; Nayga, 1998; Sabelhaus et al., 2013; Skinner, 1985). Many of the studies reviewed had an emphasis on the budgets related to raising children. Hargunani et al. (2024) analyzed family spending patterns in Mumbai and concluded that many families focus their expenditures on the current and future wellbeing of their children. This is evidenced by money spent on basic necessities and setting aside money for the future.

The United States Department of Agriculture (USDA) has produced reports that use the CEX to specifically analyze the costs of raising a child in the United States. The most recent report (Lino et al., 2017) found the top expenditure for married-couple families with two children to be housing. The rankings of other expenditures were different depending on the age of children, but food, child care/education, and transportation were always the next highest expenditures on children. Similar to the USDA report on the cost of raising children, Osborne et al. (2021) modeled the cost of raising children in Texas by following similar methodologies but using Texas-specific data for housing and childcare costs. They looked not only at married-couple families, but also at single-parent households and duel households where children spend time with both parents in different locations. They found differing expenditures on children among the different family make-ups and among different incomes.

Duncan et al. (2023) performed a study in Canada using the country's Survey of Household Spending (SHS) to analyze family expenditures. They found similar results as previously mentioned studies. Different income groups had different amounts allocated to children, but housing was always the highest expenditure with food, child care/education, and transportation being the next highest expenditures.

Other studies with similar analyses have had similar findings. Hastings (2022) used the CEX to compare expenditures between different racial and ethnic groups. When controlling for both family characteristics and income, he found that there was not a significant difference in total expenditures on children among racial and ethnic groups. This suggests that income and family characteristics play a larger role in family budgeting than race and ethnicity.

2.3 Household Transportation Expenses and Budgets

There have been many studies on family budgets and transportation expenses (Blumenberg, 2003; Choo et al., 2007; Ferdous et al., 2010; Haas et al., 2008; Hong et al., 2005; Morris & Wigan, 1979; Thakuriah (Vonu) & Liao, 2006).

One study focused on transportation budgets was done by Thakuriah & Liao (2005). Using CEX data, they made multiple models to analyze the expenditures related to vehicle ownership of households in the United States. In each model, they used a variety of variables (income, household demographics, spatial factors, economic factors, and family condition factors) to predict the amount of money a household spends on vehicles. Their model results indicate 18 percent of additional household expenditures is a vehicle expense. They results also indicate many factors influence household vehicle expenses. The models showed that homeowners spend more on vehicle expenses. They also showed that vehicle expenses are connected with the sex of the head of household and the number of people in the household.

In a study by Deka (2015), they used two years of the CEX to see the connection between housing expenses and transportation expenses. They created three least squares models: one to describe expenditures in dollar amounts, one to describe expenditures as percentages of income, and one to describe expenditures as shares of total household expenditures. They found a positive association between transportation expenses and housing expenses. Similarly, share of income on transportation was positively affected by share of income on housing, but there was no significant evidence showing the share of income on housing being affected by share of income on transportation. Both housing and transportation expenditures were found to be higher for those living in single-family detached homes and lower for those living in older homes.

Mattson & Peterson (2019) looked at how density, land use, transportation, and household expenditures are related. They developed a regression model using CEX data where transportation expenditures were estimated using type of housing, population, family characteristics, and other factors. Similar to other studies, they found that income is positively associated with transportation expenditures. Their model showed single-family homes to have higher transportation expenditures than other types of dwellings. Using data from the U.S. Census Bureau's Annual Survey of State and Local Government Finances, they also created models to explore how land use might affect municipal spending. In these models, increase in density was associated with a decrease in many per capita operational costs, including fire protection, streets and highways, parks and recreation, and others.

A similar study was published by Mattson (2020) where he modeled transportation expenditures using CEX data. The results showed, like the previous publication, that single-family dwellings spend more on transportation compared to other types. He also found that larger household sizes and newer homes contribute to higher transportation spending.

In their study, Molloy et al. (2024) look at CEX data after proposing a new framework to look at transit users and vehicle users. In past studies, three categories have been used to analyze transportation users: "drivers" are those who own a car, "captive riders" are those who use transit because they ca not afford a car, and "choice riders" are those who can afford a car but do not. Instead of having one grouping for those that own cars, Molloy et al. (2024) propose splitting that category into "captive drivers" and "choice drivers". Captive drivers would be those in the population that are low income but have a car representing people with less transportation freedom. They analyzed the CEX with this new way of classifying transportation users and found that captive drivers carry the most transportation burden. The transportation expenditures of captive drivers average more than 16% of total household expenditures while the transportation expenditures of captive riders was only around 7% of total expenditures.

2.4 The Current Study

Like many of the existing studies described, the current study improves our understanding of household spending patterns and transportation choices. It adds to the literature in three main ways: Dataset, Methodology, and Research Question.

2.4.1 Dataset

Many studies seek to understand household transportation choices with time use diaries or travel surveys. This study, on the other hand, seeks to answer similar questions using expenditure data. Instead of using day to day activities to understand where and how a household is getting around, we use households' weekly expenses to find underlying patterns that explain, and perhaps shape, their travel choices.

Another important thing to note is that all of the reviewed transportation studies using the CEX were using data from before 2020. Since much has changed since the COVID19 pandemic, the 2024 data used in the current study will provide a more timely analysis.

2.4.2 Methodology

The latent class analysis for the current study looks at unique exogenous and outcome variables. To perform the analysis, we create

2.4.3 Research Question

The purpose of the current study is to see how household budgets are structured around transportation decisions. We are going about it with the perspective that transportation expenditures help define a household. Looking at the data in this way will help us…

3 Data

Data for the following analysis were attained from the 2024 Consumer Expenditure Survey (CEX) Public Use Microdata (PUMD). The CEX is administered continuously by the Bureau of Labor Statistics (BLS). The data are organized by quarter and published each year. The CEX consists of two surveys as described below.

- **The Interview Survey** tracks a household's major and recurring expenditures by reviewing their expenses from the past three months. Approximatedly 6000 usable interviews are collected each quarter. After interviewing a household for 4 consecutive quarters, they are removed from the sample and replaced by a new household.
- **The Diary Survey** tracks a household's weekly expenditures by having them fill out a detailed expenditure diary across a two-week span. Approximately 3000 usable responses are collected each quarter. After the two consecutive weeks of reporting expenditures, the household is removed from the sample and replaced by a new household.

This study uses data from the Diary Survey.

Source: Article Notebook

Of the 397 different types of expenditures found in the data, we created 24 categories: Air Travel, Alcohol, At -Home Entertainment, Bike or Scooter or other Single Rider Travel, Bus Travel, Car Fuel, Car Maintenance, Car Other, Car Payment, Car Purchase, Clothing Related, Education, Family Spending, Food Related, House Related, Hygiene, Medical, Motorcycle Travel, no category specified, Recreational Related, School Supplies, Ship Travel, Taxi or Limo Travel, Train Travel.

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